Kuramoto dynamics on a dumbbell graph: reduction to the hyperbolic disk

Mathematical notes for the `kuramoto-visuals` repository

Supersedes the formulas of the bachelor thesis `v0.pdf`

Abstract

These notes fix the mathematics behind the code in this repository. They start from the Kuramoto system on a dumbbell graph (two complete graphs joined by a chain), rewrite it in complex form, reduce each complete-graph component to the Möbius group by the Watanabe–Strogatz ansatz, fix the gauge by the conformal barycenter, and end with the variables and formulas that are actually integrated and drawn. The starting point is the bachelor thesis kept next to this file as `v0.pdf`; where a formula below differs from the thesis this is said in a remark. The formulas are backed by numerical tests; section 9 lists which.

Contents

1 Setting **2**

2 Complex form **3**

3 The complete graph **3**
3.1 Möbius action and the Watanabe–Strogatz ansatz . . . 3
3.2 The closed equation for w . . . 4
3.3 Gradient flow in the hyperbolic metric . . . 4
3.4 First integrals . . . 5

4 The dumbbell: reduction of the cliques **5**
4.1 The reduced system . . . 5
4.2 The form with ψ . . . 5
4.3 The potential is no longer a Lyapunov function . . . 6

5 Gauge and frames **6**
5.1 Gauge freedom . . . 6
5.2 Conformal barycenter . . . 6
5.3 Canonical gauge . . . 7
5.4 Three frames . . . 7
5.5 The global center . . . 7

6 Exponential coordinates **8**

7 Numerically stable forms **9**

8 Linear analysis and time scales **10**
8.1 Energy . . . 10
8.2 Synchrony and the slow time scale . . . 10
8.3 The collapsed-cluster system . . . 10
8.4 Stationary states . . . 11
8.5 Two reference states . . . 11

9 Formulas and code **11**

A Proof of Proposition 1 **12**

B Derivation of (39) **13**

C Derivation of (27) **13**

1 Setting

The graph has $2N + l$ vertices: two complete graphs on N vertices and a chain of l vertices between them. Vertices are numbered

$$\underbrace{1,\dots,N}_{\text{first clique}},\qquad \underbrace{N+1,\dots,N+l}_{\text{chain}},\qquad \underbrace{N+l+1,\dots,2N+l}_{\text{second clique}},$$

and the chain is attached to vertex N of the first clique and to vertex $N+l+1$ of the second. We call these two the *attached nodes*; the other $N-1$ vertices of a clique are its *identical nodes*.

We consider the Kuramoto model [1] on this graph. All oscillators have the same natural frequency, which is removed by passing to a rotating frame, and there is no noise. With symmetric weights a_{ij} the system is

$$\dot\theta_i = \sum_j a_{ij}\,\sin(\theta_j-\theta_i),\qquad a_{ij}=\begin{cases}\lambda/N, & i\neq j \text{ in the same clique},\\ \kappa, & i,j \text{ adjacent on the path } N,\,N+1,\dots,N+l+1,\\ 0, & \text{otherwise.}\end{cases} \tag{1}$$

Written out,

$$\begin{aligned}
\dot\theta_j &= \frac{\lambda}{N}\sum_{i=1}^{N}\sin(\theta_i-\theta_j), && j=1,\dots,N-1,\\
\dot\theta_N &= \frac{\lambda}{N}\sum_{i=1}^{N-1}\sin(\theta_i-\theta_N)+\kappa\sin(\theta_{N+1}-\theta_N),\\
\dot\theta_j &= \kappa\sin(\theta_{j-1}-\theta_j)+\kappa\sin(\theta_{j+1}-\theta_j), && j=N+1,\dots,N+l,\\
\dot\theta_{N+l+1} &= \kappa\sin(\theta_{N+l}-\theta_{N+l+1})+\frac{\lambda}{N}\sum_{i=N+l+2}^{2N+l}\sin(\theta_i-\theta_{N+l+1}),\\
\dot\theta_j &= \frac{\lambda}{N}\sum_{i=N+l+1}^{2N+l}\sin(\theta_i-\theta_j), && j=N+l+2,\dots,2N+l.
\end{aligned} \tag{2}$$

The system is invariant under a common shift of all phases.

For a clique we use its own order parameter, the centroid of its N points on the unit circle:

$$r=\frac{1}{N}\sum_{j\in\text{clique}}e^{i\theta_j},\qquad 0\le|r|\le 1. \tag{3}$$

Both cliques are treated identically, so from section 3 on we write formulas for the first one, with identical nodes $1,\dots,N-1$, attached node N and chain neighbor $N+1$. For the second clique replace these by $N+l+2,\dots,2N+l$, by $N+l+1$ and by $N+l$.

2 Complex form

Put $z_j = \mathrm{e}^{\mathrm{i}\theta_j}$. A phase equation driven by a complex field H,

$$\dot\theta = \operatorname{Im}(H\,\overline{z}) \qquad\Longleftrightarrow\qquad \dot z = \tfrac{1}{2}(H - \overline{H}\,z^2), \tag{4}$$

is a Riccati equation for z; this follows from $\dot z = \mathrm{i}z\dot\theta$ and $\operatorname{Im}(H\overline{z}) = (H\overline{z} - \overline{H}z)/2\mathrm{i}$. Since $\sum_i a\sin(\theta_i-\theta) = \operatorname{Im}(a\sum_i z_i\,\overline{z})$, the system (2) becomes

$$\dot z_j = \frac{\lambda}{2}(r - \overline{r}\,z_j^2), \qquad j = 1,\dots,N-1, \tag{5}$$

$$\dot z_N = \frac{\lambda}{2}(r - \overline{r}\,z_N^2) + \frac{\kappa}{2}(z_{N+1} - \overline{z}_{N+1}\,z_N^2), \tag{6}$$

$$\dot z_j = \frac{\kappa}{2}\Big[(z_{j-1}+z_{j+1}) - \overline{(z_{j-1}+z_{j+1})}\,z_j^2\Big], \qquad j = N+1,\dots,N+l, \tag{7}$$

with r the order parameter (3) of the first clique, which includes the attached node z_N. In (5) the self-term of the sum cancels, so it may be included.

The point of (5) is that the $N-1$ identical nodes obey the *same* Riccati equation with the same field $H = \lambda r(t)$. Nothing else about r is used in the reduction: it may depend on time in any way.

3 The complete graph

In this section $\kappa = 0$ and we consider one clique alone, so that all N nodes obey (5). This is the classical case [2, 3, 4].

3.1 Möbius action and the Watanabe–Strogatz ansatz

For $p \in \mathbb{D} = \{|z| < 1\}$ let

$$T_p(z) = \frac{z-p}{1-\overline{p}\,z}, \qquad T_p^{-1} = T_{-p}, \qquad T_p(p) = 0. \tag{8}$$

Every conformal automorphism of the disk is $z \mapsto \zeta\,T_w(z)$ with $|\zeta| = 1$, $w \in \mathbb{D}$; these maps preserve the unit circle and form a three-parameter group isomorphic to $PSU(1,1) \cong PSL(2,\mathbb{R})$, the group of orientation-preserving isometries of the hyperbolic disk.

A Riccati equation (4) is the infinitesimal form of such a map. Hence points driven by a common field move by a common automorphism, and one looks for the solution in the form

$$z_j(t) = \zeta(t)\,T_{w(t)}(\beta_j) = \zeta\,\frac{\beta_j - w}{1-\overline{w}\,\beta_j}, \qquad |\zeta| = 1, \quad |w| < 1, \quad |\beta_j| = 1, \tag{9}$$

with constants β_j.

Proposition 1. *Let $H(t)$ be any complex function. The functions (9) solve $\dot z_j = \frac{1}{2}(H - \overline{H}z_j^2)$ for every constant β_j on the unit circle if and only if*

$$\dot w = -\tfrac{1}{2}\,(1-|w|^2)\,\overline{\zeta}\,H, \tag{10}$$

$$\dot\zeta = -\tfrac{1}{2}\,(\overline{w}\,H - w\,\overline{H}\,\zeta^2). \tag{11}$$

The proof is a direct computation, given in Appendix A. With $H = \lambda r$ these are the equations (1.8) of the thesis.

Meaning of the variables.

- ζ is a rigid rotation of the whole configuration. We write $\zeta = \mathrm{e}^{\mathrm{i}\alpha}$.
- w measures how strongly the configuration β is squeezed. The map T_w pushes boundary points away from $w/|w|$ and towards $-w/|w|$; as $|w| \to 1$ every $\beta_j \neq w/|w|$ is sent to $-w/|w|$. Full synchronization of the clique is $|w| \to 1$.
- β_j are constants of the motion. They are not unique: the triple (w, ζ, β) has a three-parameter gauge freedom, fixed in section 5.

3.2 The closed equation for w

Insert the ansatz into the order parameter:

$$q := \overline{\zeta}\, r = \frac{1}{N}\sum_{j=1}^{N} T_w(\beta_j), \qquad |q| = |r|. \tag{12}$$

Then (10), (11) with $H = \lambda r$ read

$$\dot{w} = -\frac{\lambda}{2}\,(1 - |w|^2)\, q, \tag{13}$$

$$\dot{\alpha} = -\lambda\, \mathrm{Im}(\overline{w}\, q). \tag{14}$$

The first equation contains only w and the constants: the N-dimensional dynamics is reduced to a flow on the disk, and α is found afterwards by a quadrature.

Remark 1. In the thesis the $\dot{w}$ equations (2.8), (2.10), (2.18) are printed with the denominator $1 - w\beta_j$; it is $1 - \overline{w}\beta_j$. The $\dot{\zeta}$ line of (2.8) lacks the weights: by (14), $\dot{\zeta} = -\frac{\lambda}{2}\,\zeta\,(\overline{w}q - w\overline{q})$ and $\overline{w}\,T_w(\beta) - \text{c.c.} = \frac{1-|w|^2}{|1-\overline{w}\beta|^2}\,(\overline{w}\beta - w\overline{\beta})$.

3.3 Gradient flow in the hyperbolic metric

The Poisson kernel of the disk is

$$P(w, \beta) = \frac{1 - |w|^2}{|w - \beta|^2}, \qquad |\beta| = 1. \tag{15}$$

It is the factor by which T_w stretches the circle at β: $P(w, \beta) = |T_w'(\beta)|$. A direct differentiation gives the identity

$$(1 - |w|^2)\,\frac{\partial}{\partial \overline{w}} \ln P(w, \beta) = T_w(\beta). \tag{16}$$

Define the potential and the hyperbolic gradient

$$f(w) = \frac{1}{N}\sum_{j=1}^{N} \ln P(w, \beta_j), \qquad \nabla_{\mathbb{H}} = \tfrac{1}{2}\,(1 - |w|^2)^2\,\frac{\partial}{\partial \overline{w}}, \tag{17}$$

the latter being the gradient for the metric $\mathrm{d}s^2 = 4|\mathrm{d}w|^2/(1 - |w|^2)^2$. Summing (16) over j turns (13) into

$$\dot{w} = -\lambda\, \nabla_{\mathbb{H}} f(w), \qquad \frac{\mathrm{d}f}{\mathrm{d}t} = -\lambda\, |q|^2 = -\lambda\, |r|^2 \leq 0. \tag{18}$$

So f is a Lyapunov function. It tends to $-\infty$ at the boundary away from the points β_j, and its only critical point in the disk is where $\sum_j T_w(\beta_j) = 0$, the conformal barycenter of β (section 5); this is a maximum, and the flow runs from it to the boundary.

Remark 2. In the thesis the first term of (2.12) is printed with $|w - \beta_j|$; the Poisson kernel has $|w - \beta_j|^2$.

3.4 First integrals

The cross-ratio

$$(z_1, z_2, z_3, z_4) = \frac{z_1 - z_3}{z_1 - z_2} \cdot \frac{z_4 - z_2}{z_4 - z_3} \tag{19}$$

is invariant under every Möbius map, hence under the flow. Among N points moved by one automorphism there are $N-3$ independent cross-ratios; these are the $N-3$ constants of motion of Watanabe and Strogatz.

4 The dumbbell: reduction of the cliques

Now $\kappa \neq 0$. The attached node N obeys (6), not (5), so it leaves the common Möbius orbit. The $N-1$ identical nodes still obey (5) with the common field λr, and proposition 1 applies to them verbatim.

4.1 The reduced system

Describe the identical nodes by the ansatz (9) with $j = 1, \dots, N-1$ and keep the attached node as a variable. The order parameter of the clique becomes

$$q := \overline{\zeta}\, r = \frac{1}{N}\Big[\sum_{j=1}^{N-1} T_w(\beta_j) + \overline{\zeta}\, \mathrm{e}^{\mathrm{i}\theta_N} \Big], \tag{20}$$

and the equations are

$$\dot{w} = -\frac{\lambda}{2}\,(1 - |w|^2)\, q, \tag{13}$$

$$\dot{\alpha} = -\lambda\, \mathrm{Im}(\overline{w}\, q), \tag{14}$$

$$\dot{\theta}_N = \lambda\, \mathrm{Im}(r\, \mathrm{e}^{-\mathrm{i}\theta_N}) + \kappa \sin(\theta_{N+1} - \theta_N), \tag{21}$$

$$\dot{\theta}_j = \kappa \sin(\theta_{j-1} - \theta_j) + \kappa \sin(\theta_{j+1} - \theta_j), \qquad j = N+1, \dots, N+l, \tag{22}$$

together with the same four equations for the second clique, whose attached node $N+l+1$ is coupled to θ_{N+l}. In (21) $r = \zeta q$. The phases of the identical nodes are recovered from (9).

Count. Each clique contributes w (two real numbers), α and θ_N; with the chain this is $l+8$ real variables instead of $2N+l$. (The thesis counts $l+6$, taking each complex w as one.) In each clique the $N-1$ identical nodes carry $N-4$ independent cross-ratios (19), one fewer than for the isolated complete graph.

4.2 The form with ψ

The thesis keeps the attached node inside the ansatz by letting its "constant" move:

$$\psi := T_w^{-1}(\overline{\zeta}\, z_N), \qquad z_N = \zeta\, T_w(\psi), \qquad \psi = \mathrm{e}^{\mathrm{i}\chi}. \tag{23}$$

For $\kappa = 0$ this is one more constant. In general only the chain term moves it, divided by $\partial z_N / \partial \psi = \zeta(1 - |w|^2)/(1 - \overline{w}\psi)^2$:

$$\dot{\psi} = -\frac{\kappa}{2\,(1 - |w|^2)}\Big[\zeta\, \overline{z}_{N+1}\, (\psi - w)^2 - \overline{\zeta}\, z_{N+1}\, (1 - \overline{w}\psi)^2\Big], \tag{24}$$

or, in terms of the angle,

$$\dot{\chi} = \frac{\kappa \sin(\theta_{N+1} - \theta_N)}{P(w, \psi)}. \tag{25}$$

Equations (13), (14), (24) with $q = \frac{1}{N}\big[\sum_{j<N} T_w(\beta_j) + T_w(\psi)\big]$ are the system (2.8)–(2.10) of the thesis and are equivalent to (13)–(21).

The two forms differ in practice. As the clique synchronizes, $|w| \to 1$ and $P(w,\psi) \to 0$, so (25) is singular: ψ races towards $w/|w|$ while θ_N itself moves slowly. The code therefore integrates θ_N by (21) and never forms ψ.

4.3 The potential is no longer a Lyapunov function

With the moving point ψ the potential (17) becomes

$$f(w,\psi) = \frac{1}{N}\Big[\sum_{j=1}^{N-1} \ln P(w,\beta_j) + \ln P(w,\psi)\Big], \tag{26}$$

and by (16) the equation for w is still $\dot w = -\lambda \nabla_{\mathbb{H}} f$, now for a landscape that moves with ψ. Along solutions

$$\frac{\mathrm{d}f}{\mathrm{d}t} = -\lambda\,|r|^2 + \frac{2\kappa}{N}\cdot\frac{\operatorname{Im}(\overline{w}\,\psi)}{1-|w|^2}\cdot\sin(\theta_N - \theta_{N+1}). \tag{27}$$

The second term has no definite sign and carries the factor $(1-|w|^2)^{-1}$, so f need not decrease. The derivation is in Appendix C.

Remark 3. In the thesis the second term of (2.16)–(2.17) is printed without the factor $1/N$, which comes from the $1/N$ in front of $\ln P(w,\psi)$.

5 Gauge and frames

5.1 Gauge freedom

The ansatz (9) does not determine (w,ζ,β). For any automorphism G of the disk, the constants $G(\beta_j)$ together with the map $\zeta T_w \circ G^{-1}$ describe the same motion. The thesis and most of the literature take $w(0)=0$, $\zeta(0)=1$, $\beta_j = z_j(0)$. We use a different, canonical choice which gives w a meaning independent of the initial instant.

5.2 Conformal barycenter

For points $z_1,\dots,z_n$ on the unit circle, $n \ge 3$ distinct, the *conformal barycenter* is the unique $p \in \mathbb{D}$ with

$$\sum_{j=1}^{n} T_p(z_j) = 0, \qquad p =: \operatorname{bary}(z_1,\dots,z_n). \tag{28}$$

Existence and uniqueness hold whenever no point carries half of the total mass or more [5]. The barycenter is conformally natural: for every automorphism M of the disk

$$\operatorname{bary}\big(M(z_1),\dots,M(z_n)\big) = M\big(\operatorname{bary}(z_1,\dots,z_n)\big). \tag{29}$$

By (16) it is the critical point of $p \mapsto \sum_j \ln P(p,z_j)$.

Computation. Newton's method in the frame recentered at the current guess. With $y_j = T_p(z_j)$, $m = \frac{1}{n}\sum y_j$ and $B = \frac{1}{n}\sum y_j^2$,

$$d = \frac{m + B\,\overline{m}}{1-|B|^2}, \qquad p \leftarrow T_{-p}(d) = \frac{d+p}{1+\overline{p}\,d}, \tag{30}$$

because $\frac{1}{n}\sum T_d(y_j) = m - d + B\overline{d} + O(|d|^2)$. If $|d| \ge 1$ the step $d = m$ is taken instead.

5.3 Canonical gauge

We require $\mathrm{bary}(\beta_1, \dots, \beta_{N-1}) = 0$ and $\zeta(0) = 1$. Given the initial phases, with $z_j = z_j(0)$ for the identical nodes,

$$p_0 = \mathrm{bary}(z_1, \dots, z_{N-1}), \qquad \beta_j = T_{p_0}(z_j), \qquad w(0) = -p_0, \qquad \zeta(0) = 1, \tag{31}$$

which reproduces the initial data since $T_{-p_0} = T_{p_0}^{-1}$. In this gauge, by (29) applied to $M = \zeta T_w$ and $M(0) = -\zeta w$,

$$P := -\zeta\, w = \mathrm{bary}(z_1, \dots, z_{N-1}) \qquad \text{at every time.} \tag{32}$$

Thus P is a gauge-independent point of the disk attached to the cluster, its *center*. It starts wherever the initial configuration puts it and runs to the boundary as the cluster collapses. Moreover

$$T_P(z) = \zeta\, T_{-w}(\overline{\zeta}\, z), \qquad \text{hence} \qquad T_P(z_j) = \zeta\, \beta_j, \quad j = 1, \dots, N-1. \tag{33}$$

5.4 Three frames

The same state can be drawn in three ways. The user interface shows the first two, and the variant of the second from section 5.5; the third is the frame of the thesis.

Physical frame. The oscillators z_j themselves, and the center P inside the disk.

Centered frame. Everything mapped by T_P. The cluster center is at the origin and by (33) the cluster is the fixed constellation β rotated rigidly by ζ. All other points are seen through the same Möbius lens; when the cluster has collapsed they all sit near the single boundary point $\zeta\, \mathrm{e}^{\mathrm{i}\varphi}$, $\varphi = \arg w$.

Gauge frame. Everything mapped by $z \mapsto T_w^{-1}(\overline{\zeta} z)$,

$$T_w^{-1}(\overline{\zeta}\, z_j) = \beta_j, \qquad T_w^{-1}(\overline{\zeta}\, z_N) = \psi, \qquad T_w^{-1}(0) = w. \tag{34}$$

The constellation β does not move at all, the attached node is the point ψ of (23), and w is the image of the physical origin. This is the frame of the thesis: w moves over the landscape $f(\cdot, \psi)$ by (18).

The gauge frame is the centered frame rotated by $\overline{\zeta}$.

Potentials in the frames. Let $F(p) = \frac{1}{N} \sum_{j=1}^{N} \ln P(p, z_j)$ be the potential built from the current physical points of the clique. Since $P(p, z) = |T_p'(z)|$ and T_P inverts ζT_w up to a rotation,

$$F(P) = -f(w, \psi). \tag{35}$$

In the physical frame the part of F coming from the identical nodes has its maximum at P by construction, so that picture shows where the center is, not a descent. The descent of (18) is seen in the gauge frame.

5.5 The global center

The centered frame of a cluster is tied to one clique. After that clique has collapsed it shows the rest of the system as a single boundary point, so it says nothing about the slow stage. A frame for the whole system is obtained from the barycenter of *all* nodes,

$$P_{\mathrm{all}} := \mathrm{bary}(z_1, \dots, z_{2N+l}), \qquad \sum_{j=1}^{2N+l} T_{P_{\mathrm{all}}}(z_j) = 0. \tag{36}$$

It exists as long as fewer than half of the nodes coincide; a collapsed cluster carries the fraction $(N-1)/(2N+l) < \frac{1}{2}$. In the frame $T_{P_{\mathrm{all}}}$ the configuration is balanced at every time, so it cannot shrink to a point: when in the physical frame all phases have gathered in a short arc, this frame shows the shape of that arc magnified. There is no single Möbius map moving all nodes, so nothing is rigid in this frame, in contrast to (33).

The interface also marks the barycenter of the chain nodes alone,

$$P_{\mathrm{chain}} := \mathrm{bary}(z_{N+1}, \dots, z_{N+l}), \tag{37}$$

next to P_{all} and the two cluster centers; in a frame centered at some p each of these points is drawn at its image under T_p.

In double precision P_{all} is computed from the phases themselves and is resolved while $1 - |P_{\mathrm{all}}| \gtrsim 10^{-12}$, that is while the arc is wider than about 10^{-6}; for $N = l = 20$, $\lambda = 1$, $\kappa = 10$ this holds up to $t \approx 13\,\tau$.

6 Exponential coordinates

Near synchrony $1 - |w|$ is exponentially small, so w is a bad variable. Use the hyperbolic distance ρ from the origin and the direction φ:

$$w = \tanh\frac{\rho}{2}\, \mathrm{e}^{\mathrm{i}\varphi}, \qquad v := \rho\, \mathrm{e}^{\mathrm{i}\varphi}, \qquad \rho = 2\,\mathrm{artanh}\,|w|. \tag{38}$$

The complex number v is the coordinate of w in the exponential map of the hyperbolic disk at the origin; unlike (ρ, φ) it is smooth at $\rho = 0$. Equation (13) becomes

$$\dot v = -\lambda\Big[\frac{1+c}{2}\, q + \frac{k}{2}\, v^2\, \bar q\Big], \qquad c = \frac{\rho}{\sinh\rho}, \qquad k = \frac{1-c}{\rho^2}, \tag{39}$$

with $c \to 1$, $k \to \frac{1}{6}$ as $\rho \to 0$ and $c \to 0$ as $\rho \to \infty$. In components (Appendix B),

$$\dot\rho = -\lambda\, \mathrm{Re}(\mathrm{e}^{-\mathrm{i}\varphi} q), \qquad \dot\varphi = -\frac{\lambda}{\sinh\rho}\, \mathrm{Im}(\mathrm{e}^{-\mathrm{i}\varphi} q). \tag{40}$$

The stiff factor $(1 - |w|^2)$ is gone and ρ grows at a bounded rate.

State. The integrated state of a clique is $(\mathrm{Re}\, v, \mathrm{Im}\, v, \alpha, \theta_N)$, with right-hand side (39), (14), (21).

Late times. Define the cluster phase and the offset of the attached node from it,

$$\Theta := \alpha + \varphi + \pi = \arg P, \qquad \delta := \theta_N - \Theta. \tag{41}$$

Once the cluster has collapsed, $T_w(\beta_j) \to -\mathrm{e}^{\mathrm{i}\varphi}$ and $\mathrm{e}^{-\mathrm{i}\varphi} q \to -\frac{N-1}{N} - \frac{1}{N}\mathrm{e}^{\mathrm{i}\delta}$, so that

$$\dot\rho \to \lambda\Big[1 - \frac{1-\cos\delta}{N}\Big], \qquad \dot\Theta \to \frac{\lambda}{N}\sin\delta. \tag{42}$$

Hence $\rho = \lambda t + O(1)$, the width of the cluster decays like $\mathrm{e}^{-\lambda t}$, and the informative quantity is $\rho - \lambda t$, which converges as $\delta \to 0$.

7 Numerically stable forms

In double precision $1-|w|$ is lost at $\rho \approx 37$, $\mathrm{e}^{-\rho}$ underflows at $\rho \approx 745$ and e^{ρ} overflows at $\rho \approx 709$, while a run to full synchrony reaches ρ in the thousands. All formulas below are written in the small quantity

$$s := 1 - \tanh\frac{\rho}{2} = \frac{2\,\mathrm{e}^{-\rho}}{1+\mathrm{e}^{-\rho}}, \qquad 1-|w|^2 = s\,(2-s), \tag{43}$$

which is computed without cancellation and simply becomes 0 for very large ρ. In particular

$$\ln(1-|w|^2) = 2\Big[\ln 2 - \frac{\rho}{2} - \ln\big(1+\mathrm{e}^{-\rho}\big)\Big] \tag{44}$$

is finite for every ρ.

Möbius map of a boundary point. For $\beta = \mathrm{e}^{\mathrm{i}b}$ put $\gamma = b - \varphi$, $u = \mathrm{e}^{\mathrm{i}\gamma}$ and compute $1-u = -2\mathrm{i}\sin\frac{\gamma}{2}\,\mathrm{e}^{\mathrm{i}\gamma/2}$. Then

$$T_w(\beta) = \mathrm{e}^{\mathrm{i}\varphi}\,\frac{s-(1-u)}{(1-u)+s\,u} \;\xrightarrow[s\to 0]{}\; -\mathrm{e}^{\mathrm{i}\varphi}. \tag{45}$$

Microscope. After collapse all phases of the cluster agree to machine precision, but their arrangement is still determined. Write

$$\arg z_j - \Theta = \arg(1 - s\,x_j) =: s\,\mu_j, \qquad x_j = \frac{1+u_j}{(1-u_j)+s\,u_j}, \tag{46}$$

which follows from $T_w(\beta_j)/(-\mathrm{e}^{\mathrm{i}\varphi}) = 1 - s\,x_j$. For $s|x_j| < 10^{-8}$ one uses $\mu_j = -\operatorname{Im} x_j$. The rescaled offsets have a limit,

$$\mu_j \to -\cot\frac{\gamma_j}{2} \qquad (\rho\to\infty), \tag{47}$$

so the collapsed cluster keeps a fixed shape, shrinking like s.

Order-parameter deficit. From $|r|^2 = \frac{1}{N^2}\sum_{i,j}\cos(\theta_i-\theta_j)$,

$$1-|r|^2 = \frac{4}{N^2}\sum_{i<j}\sin^2\frac{\theta_i-\theta_j}{2}, \qquad 1-|r| = \frac{1-|r|^2}{1+|r|}, \tag{48}$$

where the differences are taken from the microscope: $\theta_i - \theta_j = s(\mu_i-\mu_j)$ between identical nodes and $\theta_N - \theta_j = \delta - s\mu_j$. For a collapsed cluster

$$1-|r|^2 = \frac{4(N-1)}{N^2}\sin^2\frac{\delta}{2}, \qquad 1-|r| \approx \frac{N-1}{2N^2}\,\delta^2, \tag{49}$$

so at late times $1-|r|$ decays at twice the rate of δ. The first formula of (48) applies to any set of phases; the interface also plots it for the l chain nodes alone, with N replaced by l.

Potential. In (26) the point ψ is not formed. Using $\ln P(w,\psi) = -\ln P\big(-w, T_w(\psi)\big)$ and $|w-\mathrm{e}^{\mathrm{i}b}| = |(1-u)-s|$,

$$f = \frac{1}{N}\Big[\sum_{j=1}^{N-1}\big(\Lambda - 2\ln|(1-u_j)-s|\big) - \Lambda + 2\ln|(1-u_N)-s|\Big], \tag{50}$$

with $\Lambda = \ln(1-|w|^2)$ from (44), $u_j = \mathrm{e}^{\mathrm{i}(b_j-\varphi)}$ and $u_N = \mathrm{e}^{\mathrm{i}(\theta_N-\alpha-\varphi-\pi)} = \mathrm{e}^{\mathrm{i}\delta}$. At late times $f \approx \frac{N-2}{N}\Lambda \to -\infty$ linearly in ρ.

8 Linear analysis and time scales

8.1 Energy

The full system (1) is the gradient flow of

$$E(\theta) = -\sum_{i<j} a_{ij}\cos(\theta_i - \theta_j), \qquad \dot\theta = -\nabla E, \qquad \dot E = -|\dot\theta|^2 \le 0. \tag{51}$$

Unlike f, the energy is a Lyapunov function of the whole system for all λ, κ. Its minimum $E = -\sum_{i<j} a_{ij}$ is full synchrony.

8.2 Synchrony and the slow time scale

Linearizing (1) at $\theta_1 = \cdots = \theta_{2N+l}$ gives $\dot\varepsilon = -L\varepsilon$ with the weighted graph Laplacian

$$L = \operatorname{diag}\Big(\sum_j a_{ij}\Big) - A, \qquad \varepsilon^{\mathsf T} L\,\varepsilon = \sum_{i<j} a_{ij}(\varepsilon_i - \varepsilon_j)^2 \ge 0. \tag{52}$$

L is symmetric and positive semidefinite; its kernel is the common phase shift, and since the graph is connected for $\lambda, \kappa > 0$ all other eigenvalues are positive. Full synchrony is therefore linearly stable. (The thesis reaches the same conclusion with Gershgorin discs.) The slowest relaxation is governed by the second eigenvalue,

$$0 = \lambda_1 < \lambda_2 \le \dots, \qquad \tau := \frac{1}{\lambda_2}. \tag{53}$$

There are thus two time scales. The cliques collapse at rate λ (section 6); afterwards the chain and the two collapsed clusters relax at rate λ_2, which for a long chain is much smaller. Table 1 compares τ with the decay time fitted to $|\theta_N - \theta_{N+l+1}|$ on the window $[2\tau, 4\tau]$ of a run of the reduced system.

N	l	λ	κ	$1/\lambda_2$	fitted τ
10	10	1	1	70.9134	70.9137
20	20	1	1	259.4064	259.4127
20	20	5	1	246.0971	246.1038
20	20	1	5	65.8902	65.9081
20	40	1	1	561.8780	561.8788
40	10	2	0.5	474.0627	474.0808

Table 1: Slow time scale: prediction $1/\lambda_2$ and fit, runs up to 5τ, initial phases uniform with seed 0. Produced by `backend/scripts/timescales.py`.

8.3 The collapsed-cluster system

The set where the identical nodes of each clique coincide, $\theta_1 = \cdots = \theta_{N-1} = \Theta_1$ and $\theta_{N+l+2} = \cdots = \theta_{2N+l} = \Theta_2$, is invariant. On it the system is exactly

$$\begin{aligned}
\dot\Theta_1 &= \frac{\lambda}{N}\sin(\theta_N - \Theta_1),\\
\dot\theta_N &= \frac{N-1}{N}\,\lambda\sin(\Theta_1 - \theta_N) + \kappa\sin(\theta_{N+1} - \theta_N),\\
\dot\theta_j &= \kappa\sin(\theta_{j-1} - \theta_j) + \kappa\sin(\theta_{j+1} - \theta_j), \qquad j = N+1, \dots, N+l,
\end{aligned} \tag{54}$$

and symmetrically at the other end: $l+4$ phases. This is what the dynamics becomes after the fast stage; the first line is the second relation of (42). Its linearization at synchrony has the same slowest rate λ_2 as the full system.

In the phase differences $\Delta_N = \theta_N - \Theta_1$, $\Delta_j = \theta_j - \theta_{j-1}$ $(N < j \le N+l)$ the first equations read

$$\begin{aligned} \dot\Delta_N &= -\lambda \sin\Delta_N + \kappa \sin\Delta_{N+1}, \\ \dot\Delta_{N+1} &= \frac{N-1}{N}\,\lambda \sin\Delta_N - 2\kappa \sin\Delta_{N+1} + \kappa \sin\Delta_{N+2}, \\ \dot\Delta_j &= \kappa\bigl(\sin\Delta_{j+1} - 2\sin\Delta_j + \sin\Delta_{j-1}\bigr), \qquad N+1 < j < N+l. \end{aligned} \tag{55}$$

Remark 4. These are the relative variables of section 4.4 of the thesis with the deviations ε_j set to zero. In (4.22) of the thesis the coefficient of $\sin\Delta_N$ in $\dot\Delta_{N+1}$ is printed as λ; it is $\frac{N-1}{N}\lambda$.

8.4 Stationary states

On the invariant set of (54) stationarity forces, from the first line, $\sin\Delta_N = 0$, then from the second $\sin\Delta_{N+1} = 0$, and so on along the chain: every difference is 0 or π. A difference equal to π is unstable: the corresponding edges form a cut of the graph on which all weights enter the Hessian of E as $a_{ij}\cos\pi < 0$, and the indicator vector of one side of the cut is a direction of negative curvature. Hence full synchrony is the only stable stationary state with collapsed clusters. This does not classify the stationary states of the full system with non-collapsed cliques; numerically none other than synchrony is ever reached from uniformly random initial phases.

8.5 Two reference states

Numerically the state at intermediate times is close to "collapsed clusters joined by an arithmetic progression along the chain". Such states are not stationary; their velocity fields are as follows (A is the step of the progression).

If the progression runs through the attached nodes and the clusters sit one step beyond them,

$$\theta_j = A \cdot \min\{\max\{j, N-1\}, N+l+2\} \quad \text{up to a constant,}$$

then

$$\dot\theta_j = \sin A \cdot \begin{cases} \lambda/N, & j \le N-1, \\ \kappa - \frac{N-1}{N}\lambda, & j = N, \\ 0, & N < j \le N+l, \\ -\kappa + \frac{N-1}{N}\lambda, & j = N+l+1, \\ -\lambda/N, & j \ge N+l+2. \end{cases} \tag{56}$$

If instead the clusters and the attached nodes sit on the ends of the chain,

$$\theta_j = A \cdot \min\{\max\{j, N+1\}, N+l\} \quad \text{up to a constant,}$$

then

$$\dot\theta_{N+1} = \kappa \sin A, \qquad \dot\theta_{N+l} = -\kappa \sin A, \qquad \dot\theta_j = 0 \ \text{ otherwise.} \tag{57}$$

These are (4.10) and (4.13) of the thesis.

9 Formulas and code

Table 2 lists where each formula is implemented and which test checks it. Backend modules are in `backend/`; "frontend" means the file `frontend/src/lib/mobius.ts`; tests are in `backend/tests/`.

Formula	Implementation	Test
(1)	`kuramoto.kuramoto_ode`, `build_dumbbell_adjacency`	reference for all others
(8)	`mobius.T`, `T_inv`	`test_inverse_and_circle_preserved`
(9), (13), (14), (21), (22), (39)	`reduced.disk_rhs`, `reduced_rhs`, `reconstruct`	`test_reduced_rhs_is_the_full_rhs`, `test_dumbbell_matches_full_system`
(16), (18)	`mobius.log_poisson`	`test_gradient_form_of_w_equation`, `test_isolated_complete_graph`
(19)	`reduced.cross_ratios`	`test_isolated_complete_graph`
(24), (25)	not used in code	`test_psi_form_matches_theta_form`
(26), (50), (44)	`mobius.potential_gauge`, `log_one_minus_w2`	`test_potential_rate_identity`
(27)	—	`test_potential_rate_identity`
(28), (29), (30)	`mobius.barycenter`	`test_barycenter_zero_mean_and_equivariance`
(31)	`reduced.calibrate`, `initial_state`	`test_calibration_reproduces_initial_phases`
(32), (33), (34), (35)	frontend `clusterCenter`, `toFrame`, `potential`	`test_barycenter_gauge_identities`, `test_frames`
(36), (37)	`reduced.global_frame`, `resolved_barycenter`	`test_dumbbell_matches_full_system`
(38), (43)	`mobius.w_from_v`, `v_from_w`, `s_of_rho`	`test_v_w_roundtrip`
(40), (42), (49)	—	`test_rho_rate`
(45)	`mobius.T_w_boundary`, frontend `tBoundary`	`test_stable_T_w_matches_naive`, `test_stable_T_w_matches_mpmath`
(46), (47)	`mobius.microscope`	`test_microscope_definition_and_limit`
(48)	`reduced.order_deficit`, `phase_deficit`	`test_dumbbell_matches_full_system`
(51)	`kuramoto.energy`	`test_energy_is_a_gradient_flow`
(52), (53)	`kuramoto.fiedler`, `reduced.fit_tau`	`test_linearization_at_synchrony_is_minus_laplacian`, `test_dumbbell_matches_full_system`
(54)	—	`test_collapsed_cluster_system`
(56), (57)	—	`test_near_observed_states`

Table 2: Formula, implementation, test.

Numerical setup. The reduced and the full system are integrated with an explicit Runge–Kutta method of order 8 (`DOP853`) at relative tolerance 10^{-11}, on a grid that is dense up to $t = 10/\lambda$ and uniform up to $t = 3\tau$ by default. For $N = l = 20$, $\lambda = \kappa = 1$ and a run up to $t = 5\tau$ the reconstructed phases agree with the full system to about 10^{-8} throughout, while ρ reaches ≈ 1300.

A Proof of Proposition 1

Differentiate $z = \zeta(\beta - w)/(1 - \overline{w}\beta)$ at constant β:

$$\frac{\dot z}{z} = \frac{\dot\zeta}{\zeta} - \frac{\dot w}{\beta - w} + \frac{\dot{\overline{w}}\,\beta}{1 - \overline{w}\beta}.$$

Let $y = \overline{\zeta} z = T_w(\beta)$, so that $\beta = (y + w)/(1 + \overline{w}y)$,

$$\beta - w = \frac{(1 - |w|^2)\, y}{1 + \overline{w}y}, \qquad 1 - \overline{w}\beta = \frac{1 - |w|^2}{1 + \overline{w}y}.$$

Substituting and multiplying by $z = \zeta y$,

$$\dot z = -\frac{\zeta\,\dot w}{1 - |w|^2} + \Big[\dot\zeta + \zeta\,\frac{w\,\dot{\overline{w}} - \overline{w}\,\dot w}{1 - |w|^2}\Big] y + \frac{\zeta\,\dot{\overline{w}}}{1 - |w|^2}\, y^2.$$

This must equal $\frac{1}{2}(H - \overline{H}\zeta^2 y^2)$ for all y on the circle, i.e. coefficient by coefficient. The constant term gives (10); the coefficient of y^2 gives its complex conjugate; the coefficient of y must vanish, which with (10) is $\dot\zeta = \zeta(\overline{w}\dot w - w\dot{\overline{w}})/(1-|w|^2) = -\frac{1}{2}(\overline{w}H - w\overline{H}\zeta^2)$, that is (11). □

B Derivation of (39)

Write $q = \mathrm{e}^{\mathrm{i}\varphi}(q_\parallel + \mathrm{i}q_\perp)$ with real $q_\parallel, q_\perp$. From (13) and $|w| = \tanh\frac{\rho}{2}$, $\mathrm{d}|w|/\mathrm{d}\rho = \frac{1}{2}(1-|w|^2)$:

$$\dot\rho = -\lambda q_\parallel, \qquad \dot\varphi = -\frac{\lambda}{2}\,\frac{1-|w|^2}{|w|}\,q_\perp = -\frac{\lambda}{\sinh\rho}\,q_\perp,$$

which is (40). Then $\dot v = \mathrm{e}^{\mathrm{i}\varphi}(\dot\rho + \mathrm{i}\rho\dot\varphi) = -\lambda\,\mathrm{e}^{\mathrm{i}\varphi}(q_\parallel + \mathrm{i}c\,q_\perp)$. Since $v^2\overline{q} = \rho^2\mathrm{e}^{\mathrm{i}\varphi}(q_\parallel - \mathrm{i}q_\perp)$,

$$\frac{1+c}{2}\,q + \frac{1-c}{2\rho^2}\,v^2\overline{q} = \mathrm{e}^{\mathrm{i}\varphi}(q_\parallel + \mathrm{i}c\,q_\perp),$$

which gives (39).

C Derivation of (27)

f depends on time through w and through $\chi = \arg\psi$. The first contribution is, by (18), $-\lambda|q|^2 = -\lambda|r|^2$. For the second, only the term $\frac{1}{N}\ln P(w,\psi)$ depends on χ, and

$$\frac{\partial}{\partial\chi}\ln P(w,\psi) = -\frac{\partial}{\partial\chi}\ln|\psi - w|^2 = -\frac{2\,\mathrm{Im}(\overline{w}\,\psi)}{|\psi-w|^2}.$$

Multiplying by (25), $\dot\chi = \kappa\sin(\theta_{N+1}-\theta_N)\,|\psi-w|^2/(1-|w|^2)$, and by $1/N$ gives the second term of (27).

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